

Supplementary Figures: Strain–Doping Coupling in Graphullerene

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Companion to the main manuscript on coupled (ϵ, δ) response in qHP C₆₀. Figures adopt the observables vocabulary of Capobianco *et al.* [1] and its Supplementary Information (Electron SI): band-edge π -DOS (SI Fig. S4, purple valence / teal conduction), inverse participation ratio IPR (SI Eq. S8), electronic coupling J (SI Eq. S11), Marcus λ (vertical–adiabatic split), and Franck–Condon

weighted density FCWD (MolFC generating-function approach, SI Sec. S11). Our tetramer Exp. 4–9 audits use the same labels at PBE+D3; absolute J and λ are *not* directly comparable to the Koopmans-compliant periodic supercells of Ref. 1, but provide internally consistent (ϵ, δ) trends. **DATA:** Fig. S5 λ panels refresh from `exp9.polaron_verification.json` after vertical SP converge; until then adiabatic IP/EA only.

[1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa,

Nano Letters **24**, 7891 (2024), key reference from refer-info.md - polaron transport mechanisms.

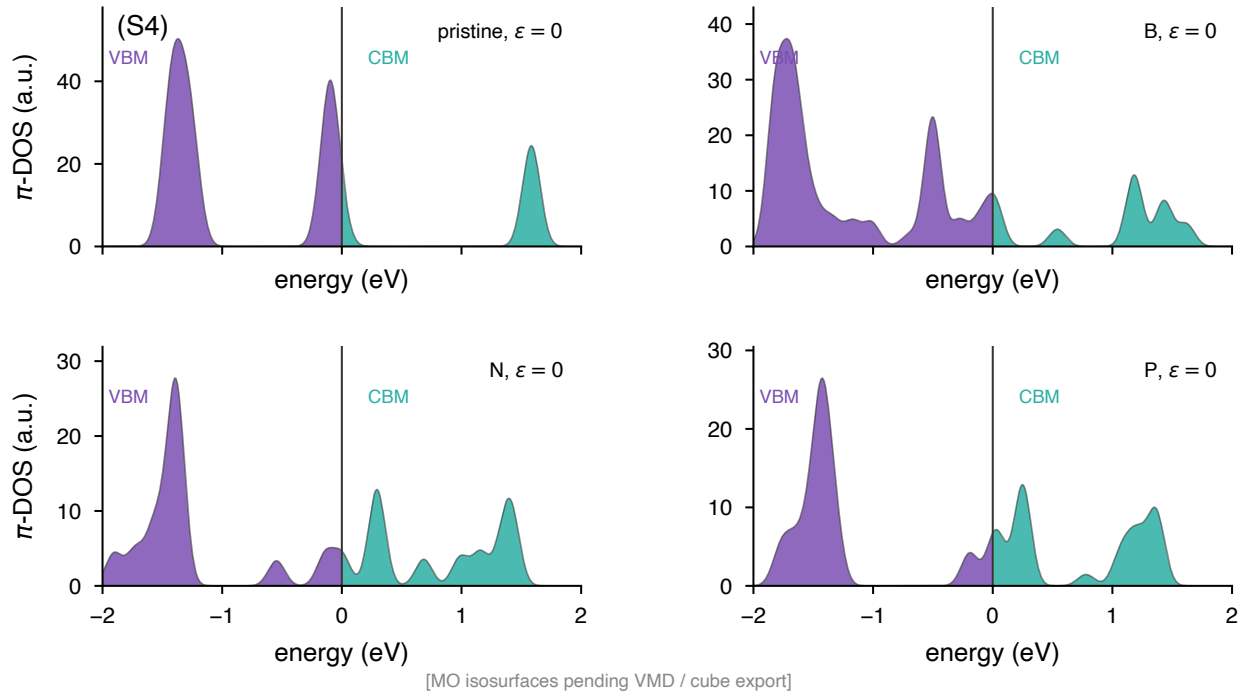


FIG. 1. π -projected DOS at $\varepsilon=0$ for pristine and B/N/P tetramers (Exp. 7 .pdos, PBE+D3). Band-edge π -DOS follows Electron SI Fig. S4 colour scheme (VBM purple, CBM teal); MO isosurfaces [pending: VMD + cube export].

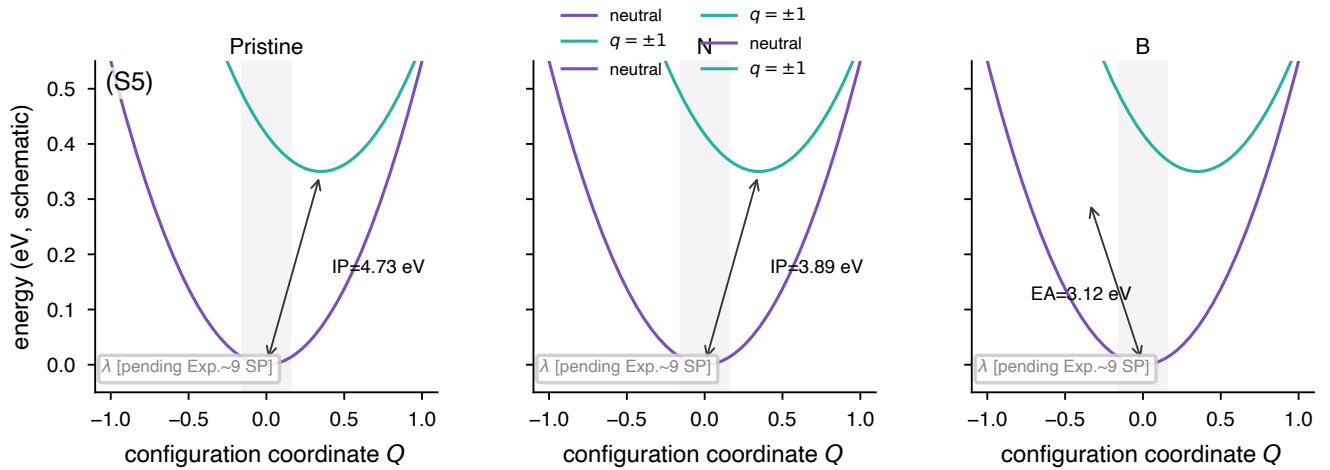


FIG. 2. Marcus configuration-coordinate schematic (Capobianco-style). Configuration-coordinate schematics following the Marcus vertical-adiabatic decomposition of Ref. 1; verified adiabatic IP/EA from converged Exp. 9 GEO.OPT pairs (pristine IP, N IP, B EA). $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ remains [pending: vertical SP at neutral geometry]—see derived.lambda.eV in audit JSON.

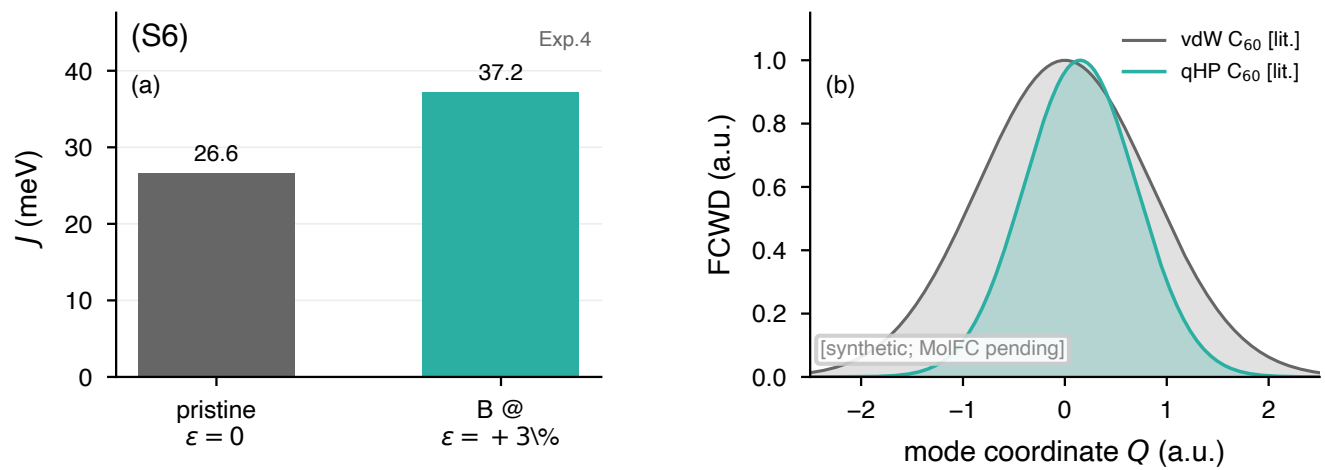


FIG. 3. Transport-adjacent observables: (a) nearest-neighbor J from Exp. 4 audit (PBE+D3, two verified tetramer points); (b) illustrative FCWD envelope (Electron SI MolFC/GF methodology [**synthetic placeholder; tetramer MolFC pending**]).